Design to Cost

Part Cost Estimator

User Guide: install it, connect to the local server, and use it

With screenshots of every main page

Screenshots use the built-in EXAMPLE rates and a demo project (SCT26 Sensor cover). Numbers shown are for illustration only, not for negotiation.

1. What this software does

Design to Cost estimates the cost of a part, its tool or mould, and the whole product BOM, from a 3D STEP model, a drawing or a bill of materials. It runs on your own computer. Nothing is sent to the internet unless you click a fetch button on the Rates page.

- Estimate: upload a STEP file and get raw material, process and part cost in seconds.
- Projects: keep parts, variants, tool cost, CAPEX plan and BOM cost against the target.
- Rates: machine-hour, material and process rates, with Draft, Reviewed and Approved versions.
- Master data: all rates and formulas are editable tables, so you can replace the examples with your own.
- Traceability: click any number to see the formula, the inputs and the rate version behind it.

IMPORTANT: The rates delivered with the software are EXAMPLES. A red banner says 'Uses example rates - not for negotiation' until you replace them (section 7).

2. How it works (30-second version)

The software is a small web server running on your laptop. You start it in a terminal window; then you use it in your normal web browser at the address http://127.0.0.1:8000. 'Connecting to the local server' simply means opening that address.

Part	What it is	Where it lives
Backend	Python (FastAPI) web server. Does all calculations.	backend/ folder
Frontend	Web pages (HTML/CSS/JavaScript). No build step.	frontend/ folder
Project database	SQLite file (projects, parts, versions). PostgreSQL optional for teams.	created on first start
Rates and master data	costing_db.sqlite and master_data.xlsx	master_data/ folder
Uploaded files	STEP files, drawings, reports	storage/ folder

3. What you need

Item	Requirement
Computer	Windows 10/11, macOS or Linux; 8 GB RAM recommended
Python	Version 3.10 or newer, from python.org (tick 'Add Python to PATH' on Windows)
Browser	Chrome, Edge or Firefox (current version)
Internet	Needed once, to install the packages. Afterwards the tool works offline.
Optional	Tesseract (reads scanned BOM drawings); Git (only if you download with git)

4. Install, step by step

Step 1. Download the software

Open https://github.com/amitkumarshukla1210/Tool_part_cost_estimator in your browser. Click the green Code button, then Download ZIP. Unzip it, for example to C:\DesignToCost. (With Git installed you can instead run: git clone https://github.com/amitkumarshukla1210/Tool_part_cost_estimator.git)

Step 2. Open a terminal in that folder

Windows: open the folder in File Explorer, click the address bar, type cmd and press Enter. Mac/Linux: open Terminal and cd into the folder.

Step 3. Create a private Python environment (once)

```
Windows:
  python -m venv .venv
  .venv\Scripts\activate

Mac / Linux:
  python3 -m venv .venv
  source .venv/bin/activate
```

The prompt now starts with (.venv). That means the environment is active.

Step 4. Install the packages (once)

```
pip install -r requirements.txt
```

This downloads about 400 MB (CadQuery / OpenCascade is the large one) and takes 3 to 10 minutes. Wait for it to finish without errors.

Step 5. Start the local server

```
uvicorn backend.app.main:app --reload
```

You should see: 'Uvicorn running on http://127.0.0.1:8000'. Leave this window open. It is the server.

Step 6. Connect: open the software in the browser

Open your browser and go to:

```
http://127.0.0.1:8000
```

The home page below should appear. If it does, you are connected.

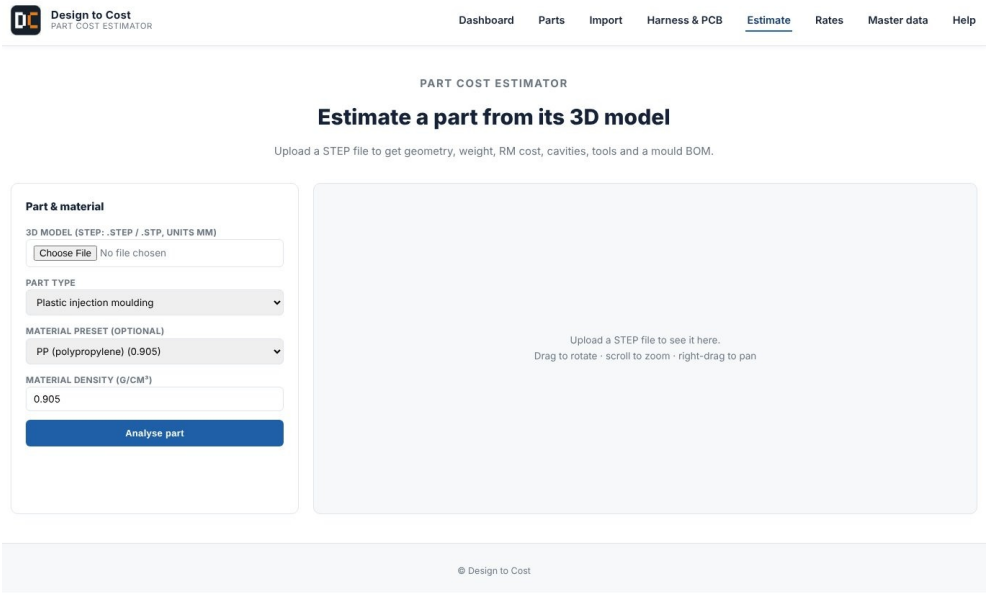


Figure 1. Home page, shown at http://127.0.0.1:8000

Stop, restart and daily use

Task	How
Stop the server	Click the terminal window and press Ctrl + C
Start again another day	Open a terminal in the folder, activate the environment (step 3, second line only), run the uvicorn command (step 5)
Use a different port	uvicorn backend.app.main:app --port 8080, then open http://127.0.0.1:8080
Let colleagues on your network use it	uvicorn backend.app.main:app --host 0.0.0.0, then they open http://<your-PC-IP>:8000. Use server mode (section 10) so they must log in.

Windows shortcut: save these three lines as start_dtc.bat in the software folder and double-click it each day.

```
@echo off
call .venv\Scripts\activate
start http://127.0.0.1:8000
uvicorn backend.app.main:app
```

5. If something goes wrong

Symptom	Fix
'python' is not recognised	Reinstall Python and tick 'Add Python to PATH'; open a new terminal. On Mac/Linux use python3.
pip install fails on cadquery	Use Python 3.10 to 3.12 (not the newest release). Upgrade pip: python -m pip install --upgrade pip, then retry.
'uvicorn' is not recognised	The environment is not active. Run the activate line (step 3) and try again.
Browser says 'can't connect'	The server window was closed or crashed. Start it again (step 5).
'address already in use'	Another program uses port 8000. Start with --port 8001 and open that address.

Page opens but looks old after an update	Hard refresh the browser: Ctrl + F5.
Scanned BOM says 'Tesseract is not installed'	Install Tesseract OCR, or upload an Excel/PDF BOM instead. Everything else works without it.

6. Guided tour with screenshots

6.1 Estimate: quick part cost from a STEP file

Click Estimate in the top menu. Choose a STEP file (samples are in the samples/ folder), pick the material, process and producing country, then Estimate.

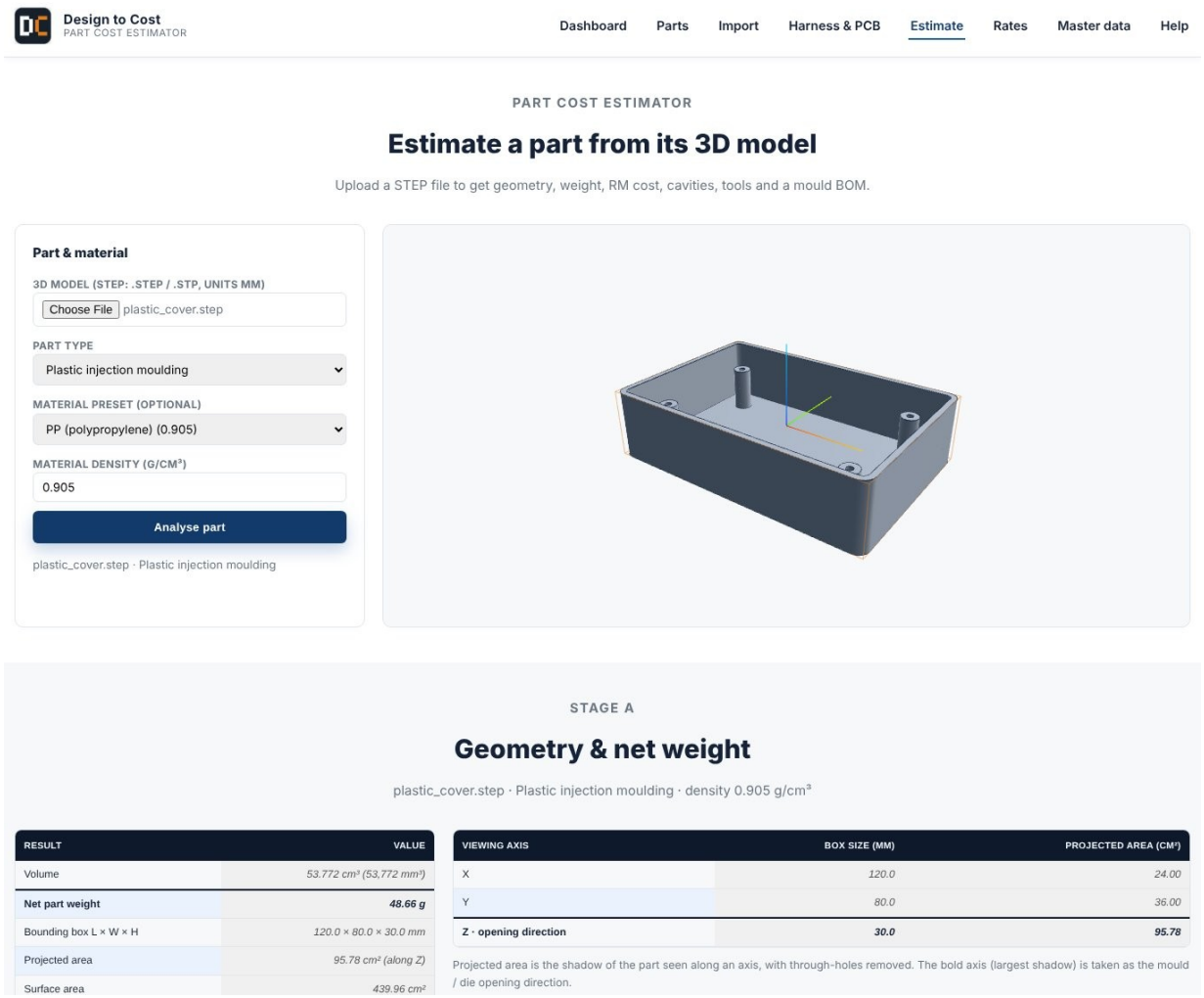


Figure 2. Estimate page: inputs and the cost result (top).

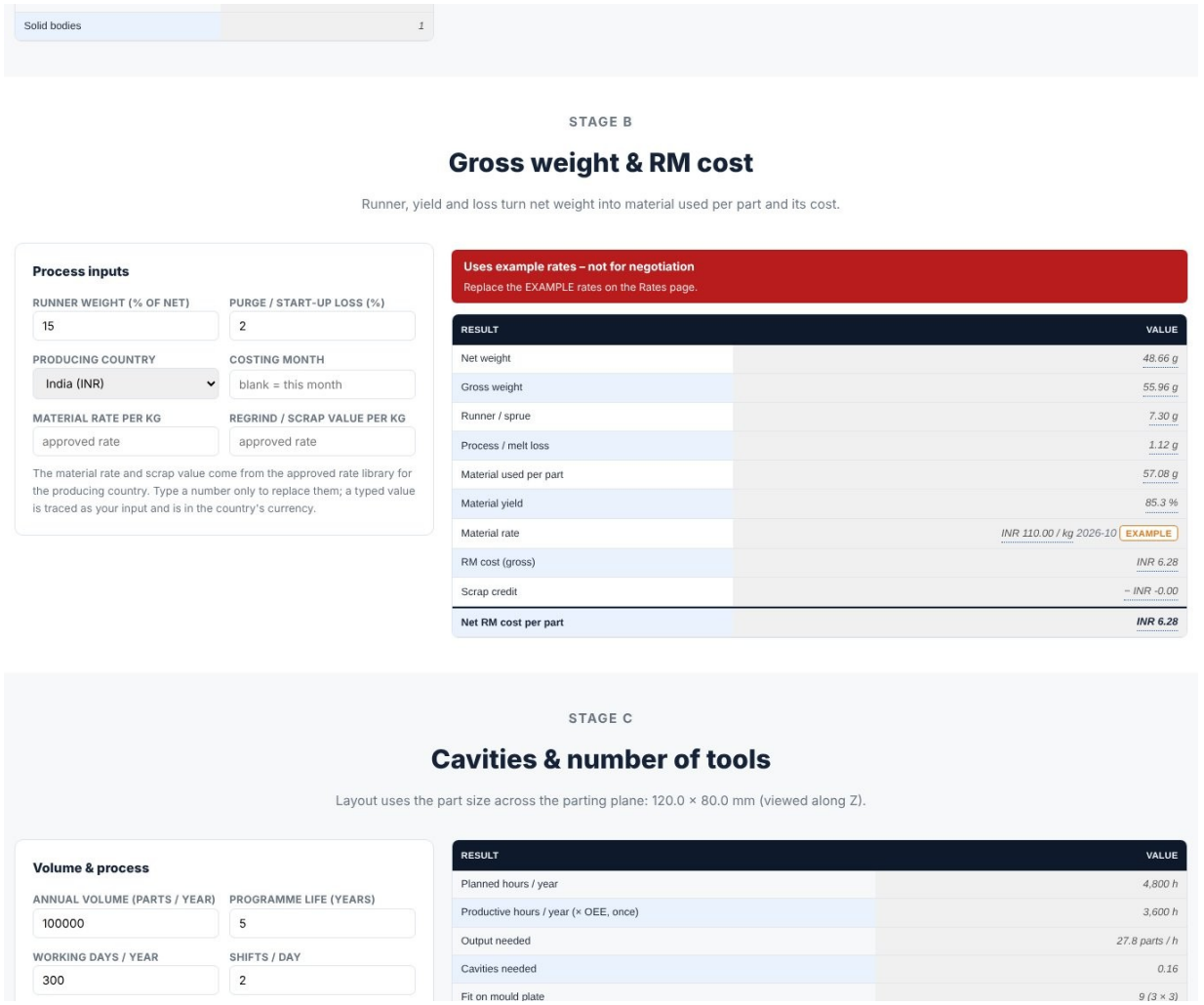


Figure 3. The result continues: cost breakdown, with every number clickable for its formula.

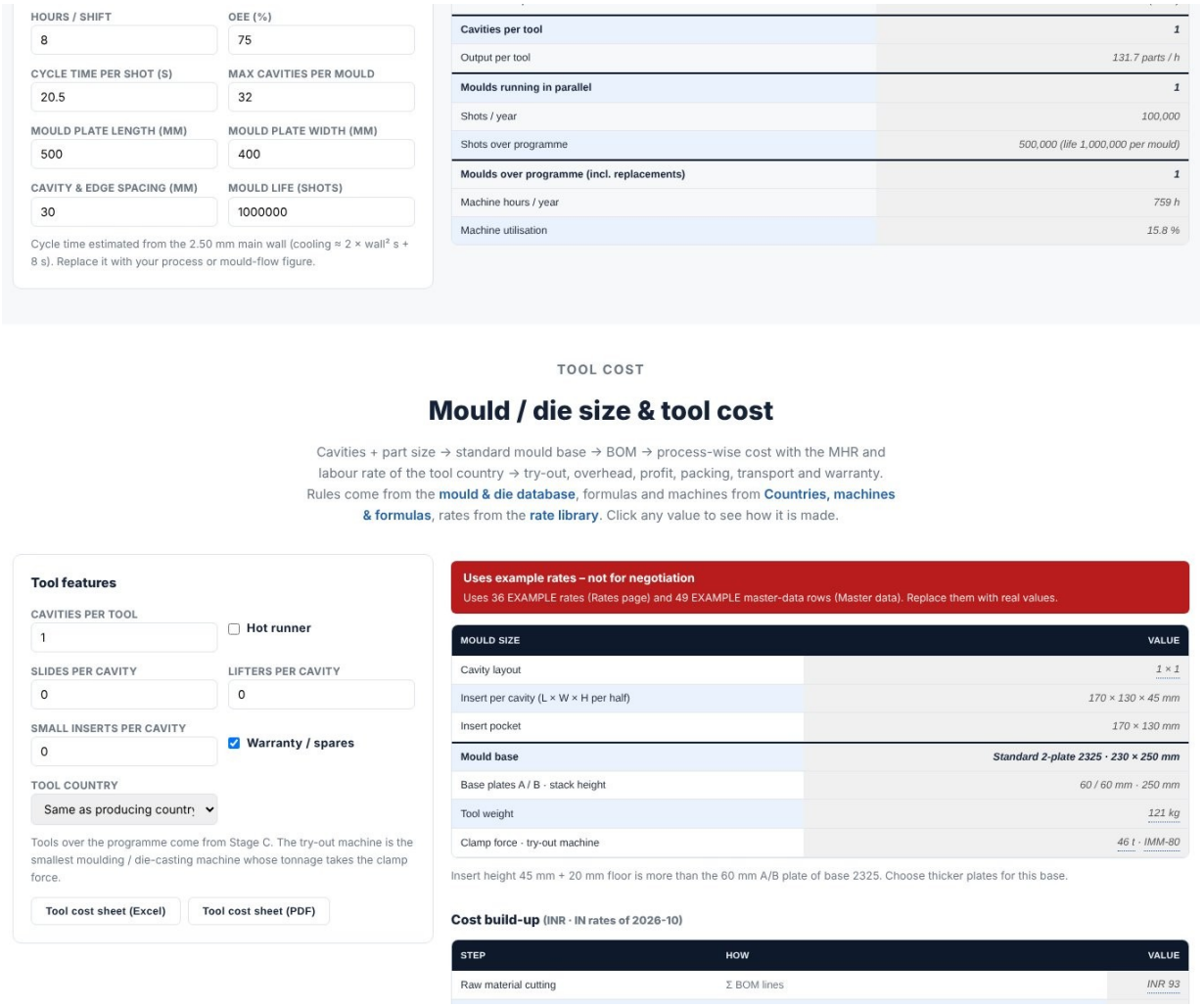


Figure 4. Result continued: process steps and rates used.

Machining (CNC, EDM, wire cut, grinding, drilling)	Σ BOM lines	INR 30,640
Polishing	Σ BOM lines	INR 1,628
Fitting & assembly	Σ BOM lines	INR 23,119
Design	Σ BOM lines	INR 8,524
Tool steel	Σ quantity × weight × steel rate	INR 6,558
Mould base & standard parts	Σ catalogue prices (INR → INR)	INR 45,860
Try-out machine & crew	trial_hours * (trial_mhr + trial_labour_per_h)	INR 6,942
Try-out material	trials * trial_shots_each * shot_weight_kg * trial_material_rate	INR 3,212
Try-out	trial_machine_cost + trial_material_cost	INR 10,154
Manufacturing cost	material_cost + bought_out_cost + process_cost + design_cost + trial_cost	INR 126,576
Toolmaker overhead	manufacturing_cost * overhead_pct / 100	INR 15,189
Toolmaker profit	(manufacturing_cost + tool_overhead) * profit_pct / 100	INR 14,176

Tool BOM & processes

Uses example rates – not for negotiation

Uses 36 EXAMPLE rates (Rates page) and 49 EXAMPLE master-data rows (Master data). Replace them with real values.

ELEMENT	CATEGORY	QTY	STEEL	STEEL COST	BOUGHT-OUT	PROCESS	HOURS / AMOUNT	PROCESS COST	LINE TOTAL
Mould base Standard 2-plate 2325 •	Bought-out	1			INR 27,300				INR 27,300
Cavity insert (steel) •	Made	1	2738 (1.2738)	INR 3,279					INR 3,279
Core insert (steel) •	Made	1	2738 (1.2738)	INR 3,279					INR 3,279
Insert machining - CNC •	Process	2				CNC milling	16.0 h	INR 11,615	INR 11,615
Insert machining - EDM •	Process	2				EDM	12.2 h	INR 7,625	INR 7,625
Insert grinding •	Process	2				Grinding	2.5 h	INR 1,079	INR 1,079
Cavity polishing •	Process	1				Polishing	5.5 h	INR 1,628	INR 1,628
Insert pocket machining •	Process	2				CNC milling	8.2 h	INR 5,975	INR 5,975
Guide pillar •	Bought-out	4			INR 5,160				INR 5,160
Guide bush •	Bought-out	4			INR 3,440				INR 3,440
Ejector pin •	Bought-out	12			INR 2,160				INR 2,160
Return pin •	Bought-out	4			INR 1,400				INR 1,400
Sprue bush •	Bought-out	1			INR 1,500				INR 1,500

• = still an EXAMPLE value in master data. Hours are priced at MHR + operators × labour rate of the tool country; per-unit processes (cutting, heat treatment, nitriding, ...) at their rate per kg / cm², at least the minimum charge per job.

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Figure 5. End of the result: totals and the traceability panel.

6.2 Dashboard: the project at a glance

The Dashboard shows BOM cost against target, CAPEX against budget, and the parts that still need attention.

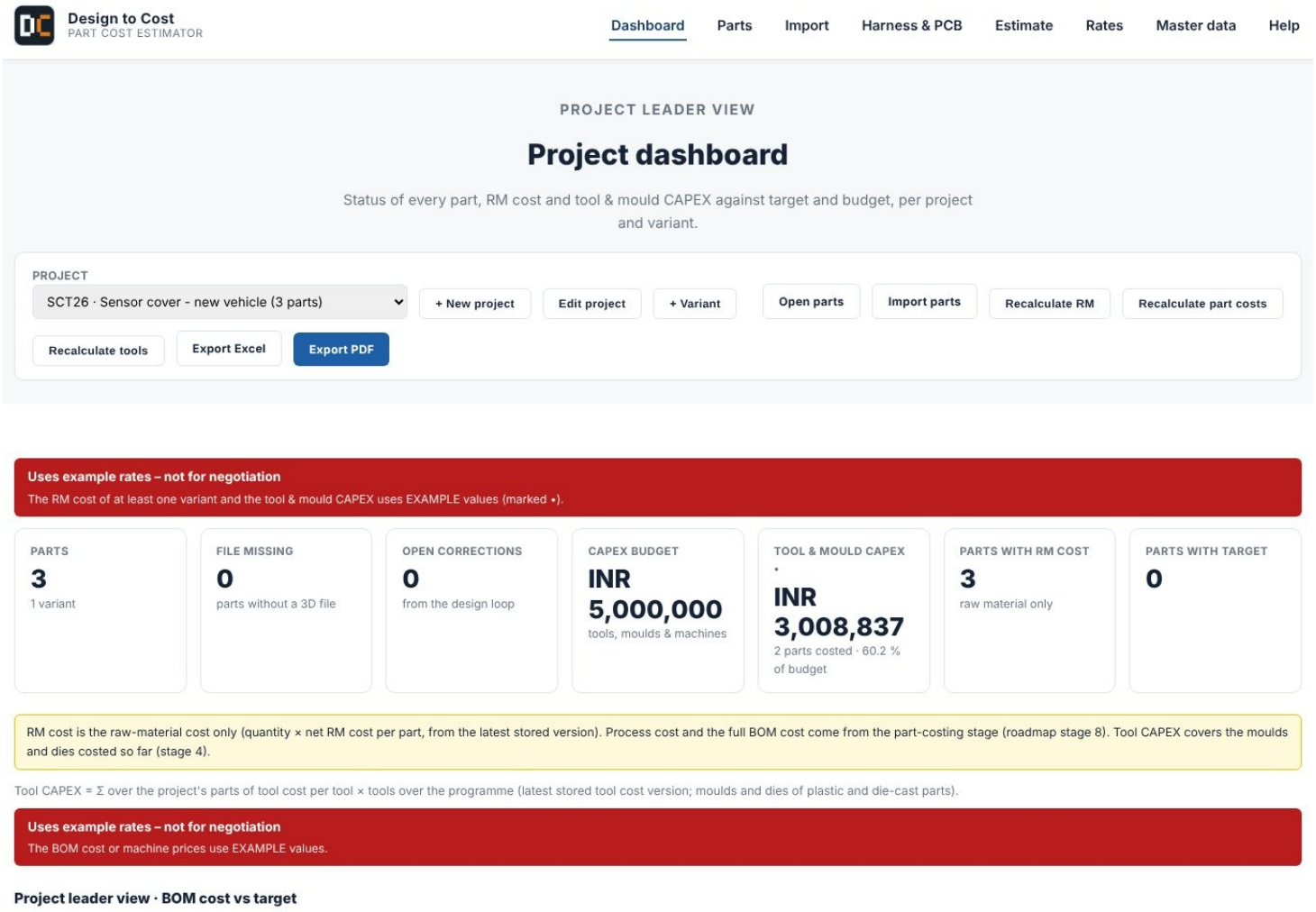


Figure 6. Dashboard: targets, BOM cost and status cards.

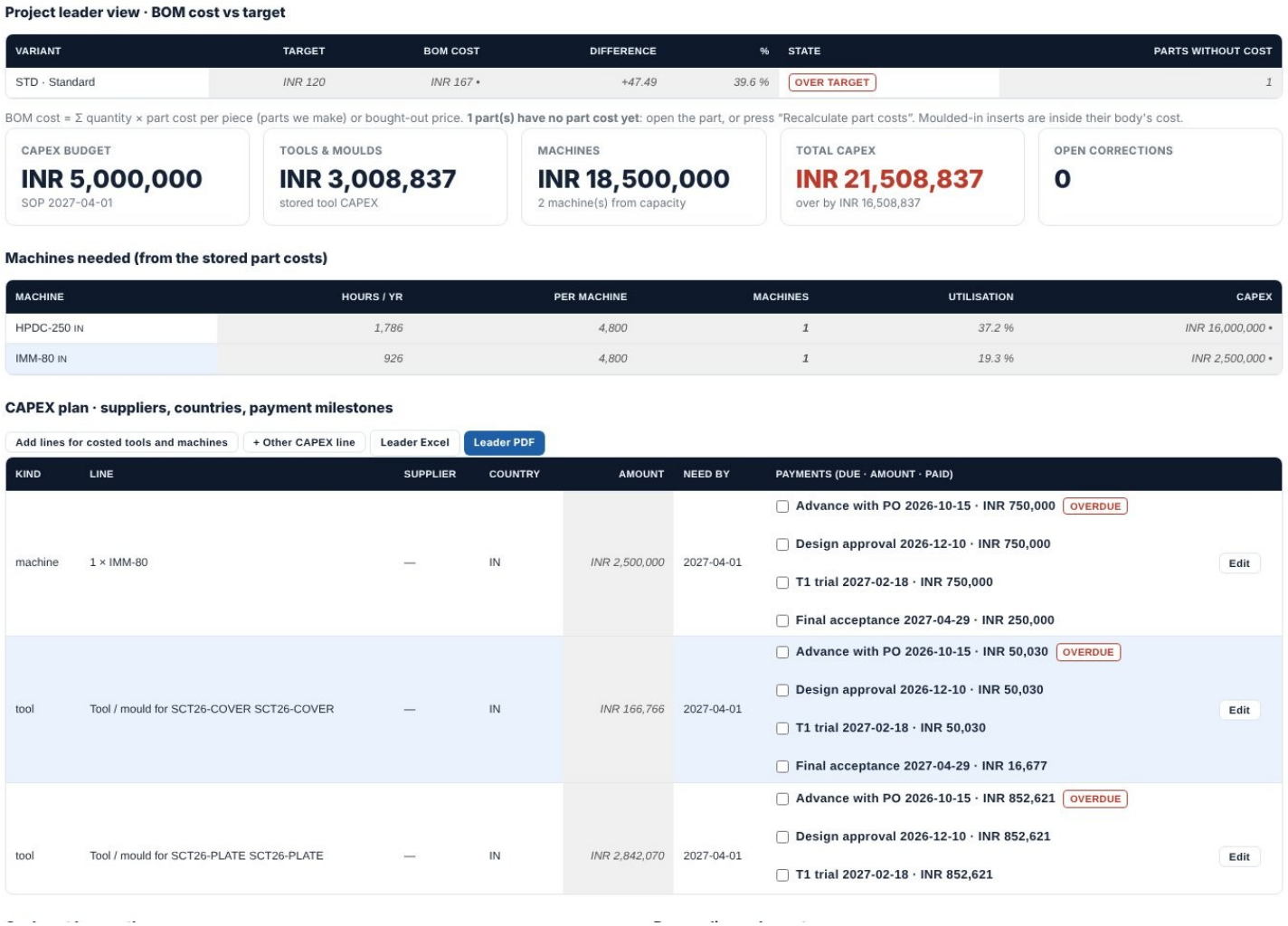


Figure 7. Dashboard: lower cards and the list of open items.

6.3 Import: bring in an assembly or BOM

Import reads an assembly STEP file, or a BOM from Excel, PDF or DXF. It shows a preview where you check every line before it is saved.

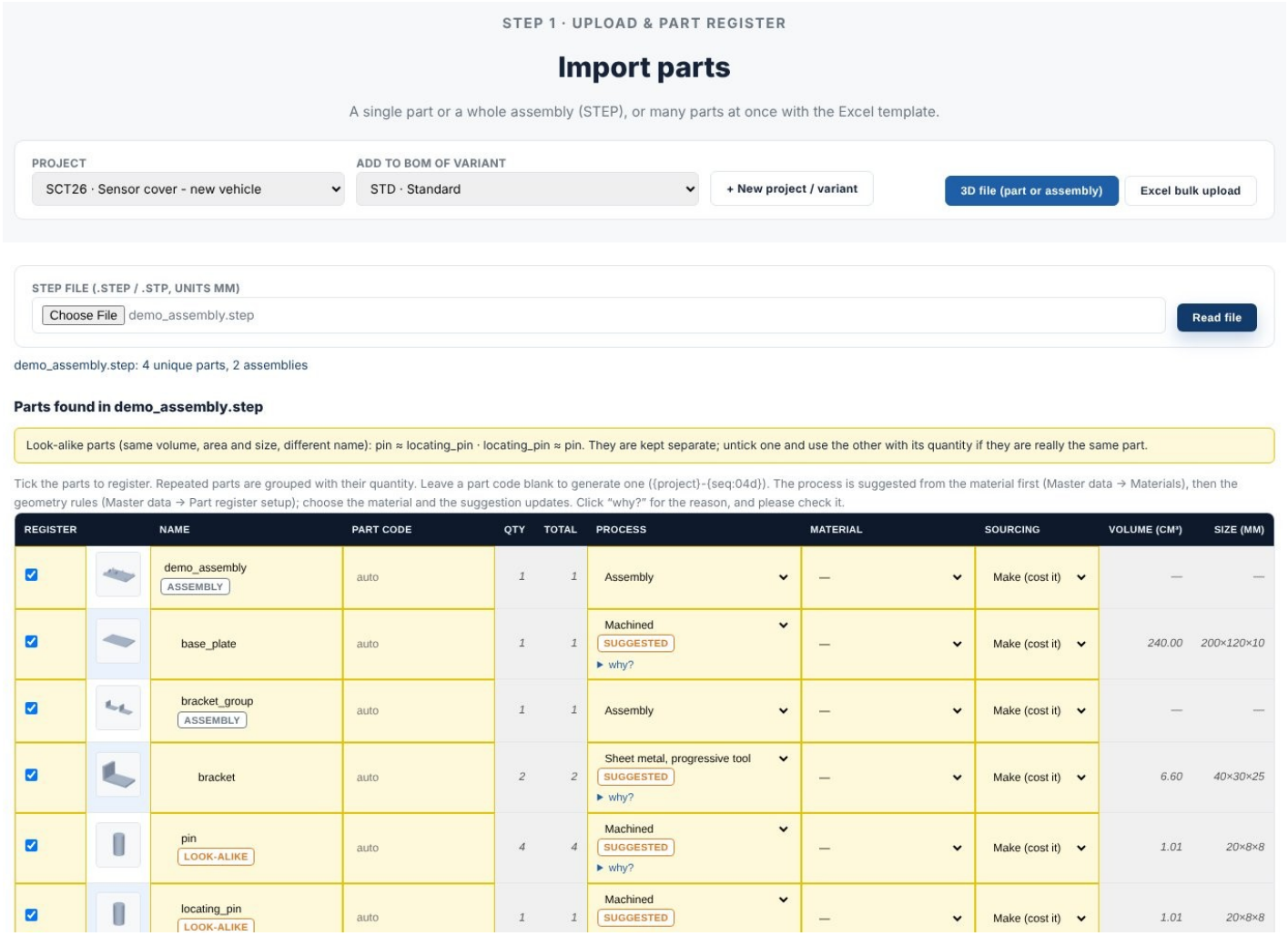


Figure 8. Import preview: check, correct, then save into the project.

6.4 Parts list and the part panel

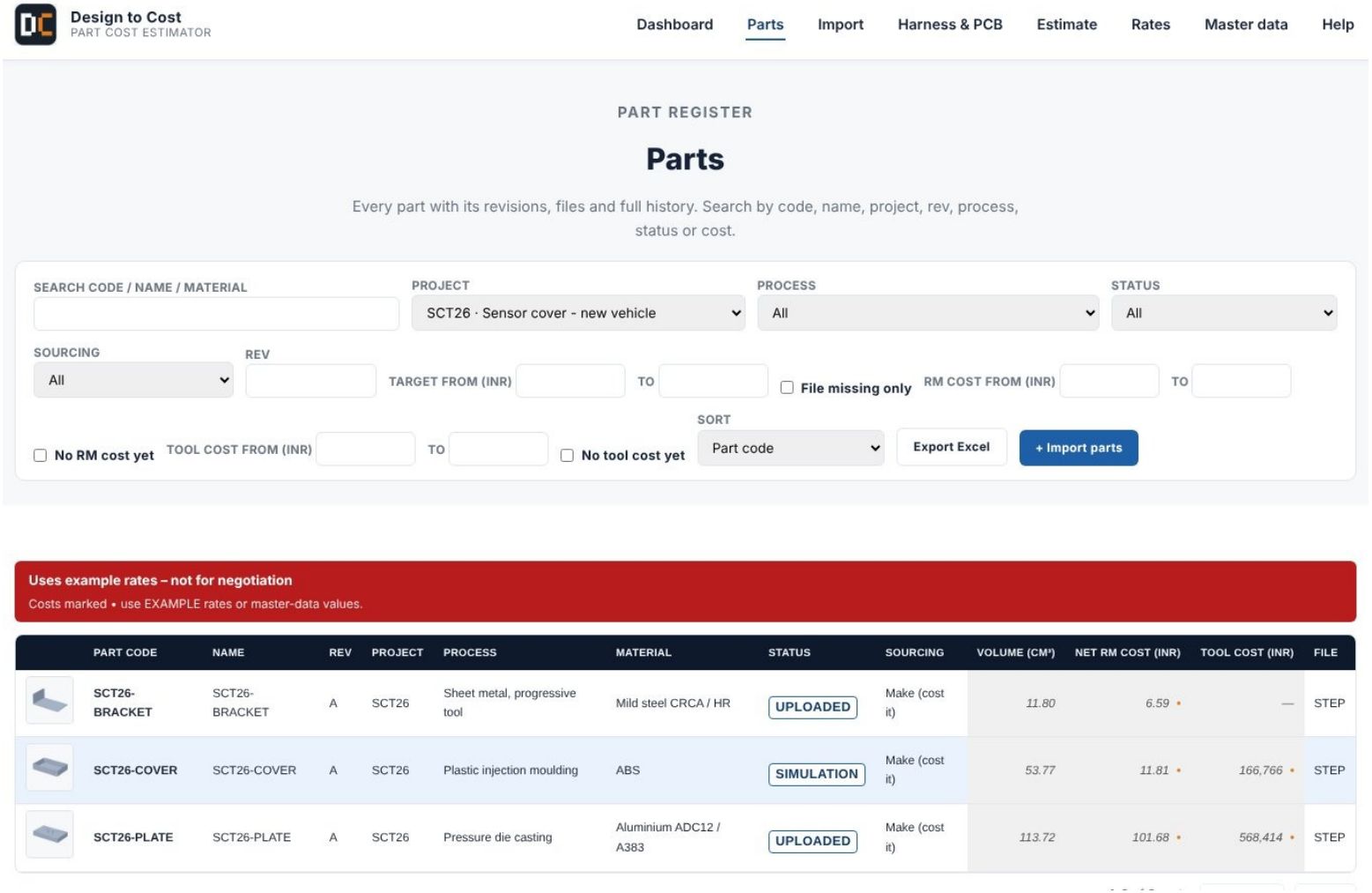
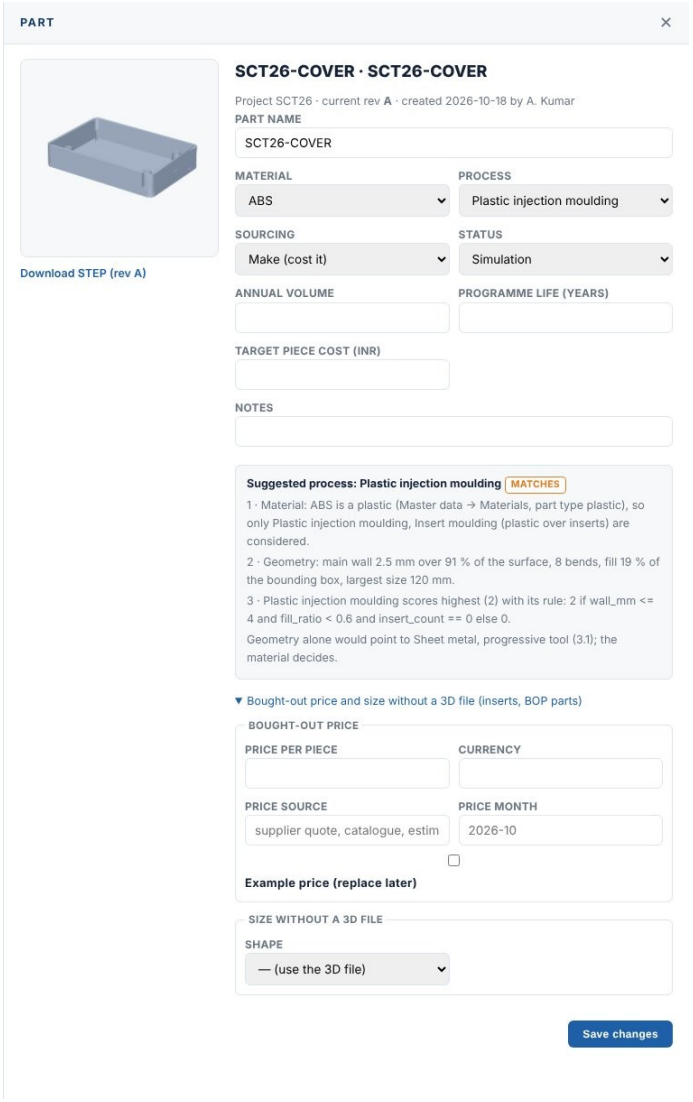


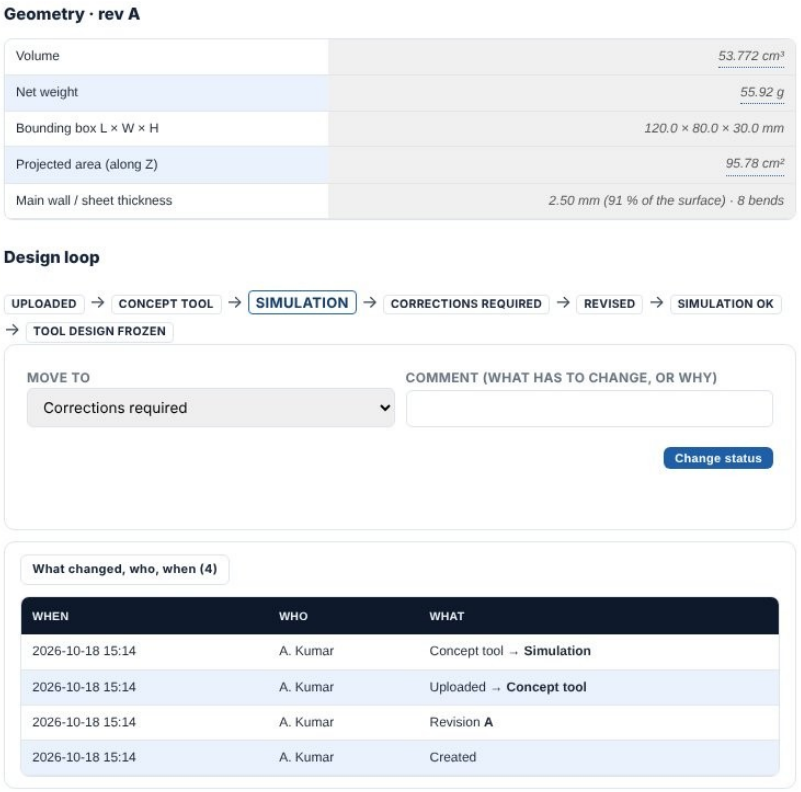
Figure 9. Parts: one row per part with material, process, tool and part cost.

Click a part to open its panel. The panel works from top to bottom. Each block uses the result of the one above.



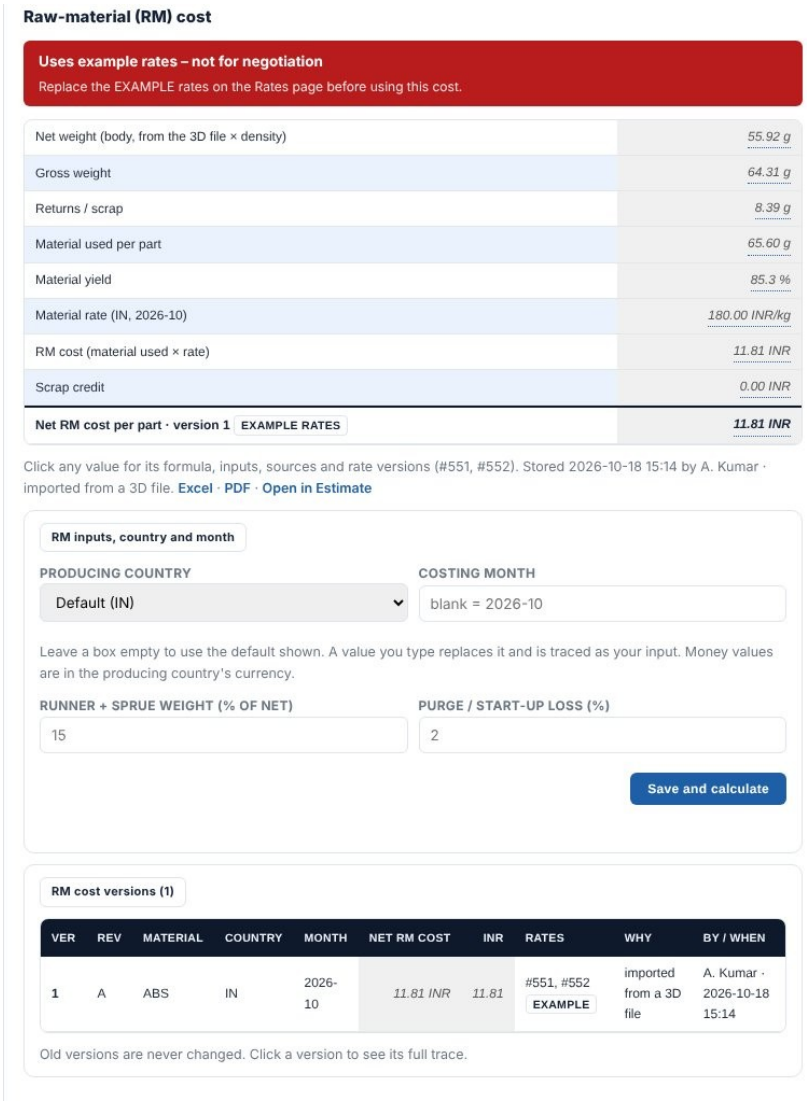
Part details

- Code, name, material, process, country, volume.
- Upload or replace the STEP model.
- Every change creates a new numbered revision. Old versions are never changed.



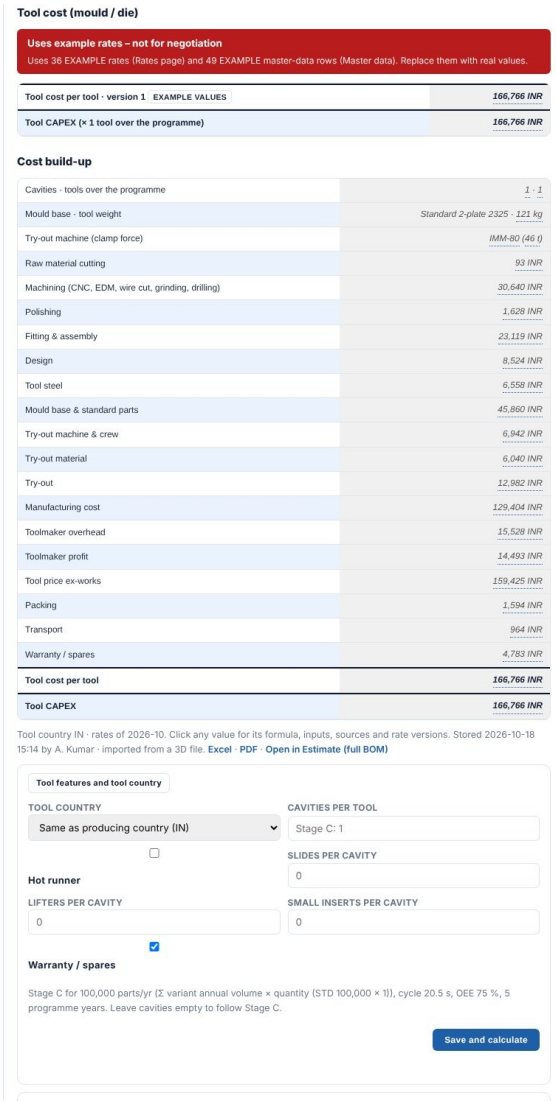
Geometry and design loop

- Volume, weight, wall thickness, bounding box read from the STEP file.
- The design loop compares the part cost to the target and says what to change.



Raw material cost

- Net weight + scrap/runner allowance, times the approved material rate.
- Click any number to see the formula.



Tool cost (mould / die)

Uses example rates – not for negotiation
Uses 36 EXAMPLE rates (Rates page) and 49 EXAMPLE master-data rows (Master data). Replace them with real values.

Tool cost per tool · version 1 EXAMPLE VALUES	***166,766 INR***
Tool CAPEX (× 1 tool over the programme)	***166,766 INR***

Cost build-up

Cavities · tools over the programme	*1 · 1*
Mould base · tool weight	*Standard 2-plate 2325 · 121 kg*
Try-out machine (clamp force)	*IMM-80 (46 t)*
Raw material cutting	*93 INR*
Machining (CNC, EDM, wire cut, grinding, drilling)	*30,640 INR*
Polishing	*1,628 INR*
Fitting & assembly	*23,119 INR*
Design	*8,524 INR*
Tool steel	*6,558 INR*
Mould base & standard parts	*45,860 INR*
Try-out machine & crew	*6,942 INR*
Try-out material	*6,040 INR*
Try-out	*12,982 INR*
Manufacturing cost	*129,404 INR*
Toolmaker overhead	*15,528 INR*
Toolmaker profit	*14,493 INR*
Tool price ex-works	*159,425 INR*
Packing	*1,594 INR*
Transport	*964 INR*
Warranty / spares	*4,783 INR*
Tool cost per tool	***166,766 INR***
Tool CAPEX	***166,766 INR***

Tool country IN · rates of 2026-10. Click any value for its formula, inputs, sources and rate versions. Stored 2026-10-18 15:14 by A. Kumar · imported from a 3D file. **Excel** · **PDF** · **Open in Estimate (full BOM)**

Tool features and tool country

TOOL COUNTRY: Same as producing country (IN)

CAVITIES PER TOOL: Stage C: 1

Hot runner

SLIDES PER CAVITY: 0

LIFTERS PER CAVITY: 0

SMALL INSERTS PER CAVITY: 0

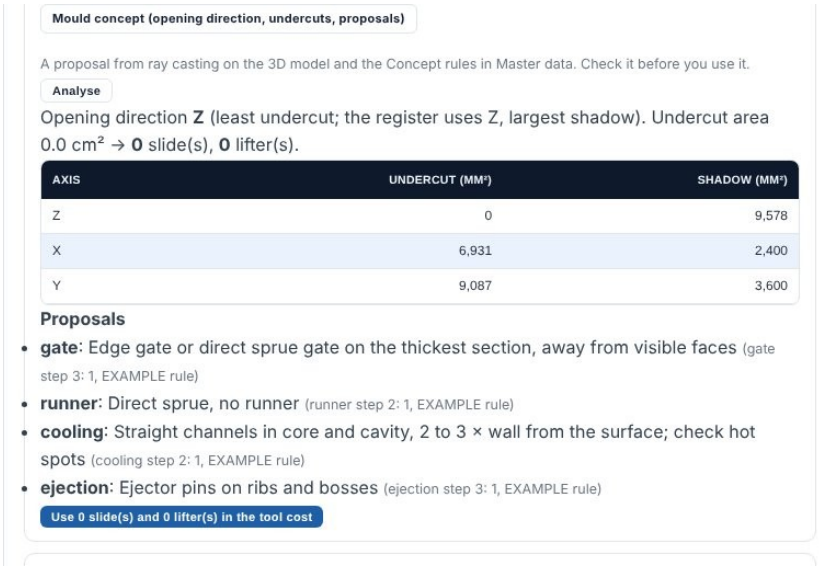
Warranty / spares

Stage C for 100,000 parts/yr (Σ variant annual volume × quantity (STD 100,000 × 1)), cycle 20.5 s, OEE 75 %, 5 programme years. Leave cavities empty to follow Stage C.

Save and calculate

Tool cost

- Choose the mould or die family (cavities, steel, size).
- Tool cost is built from steel, machining hours, standard parts and trials.
- Features such as slides and lifters add cost.



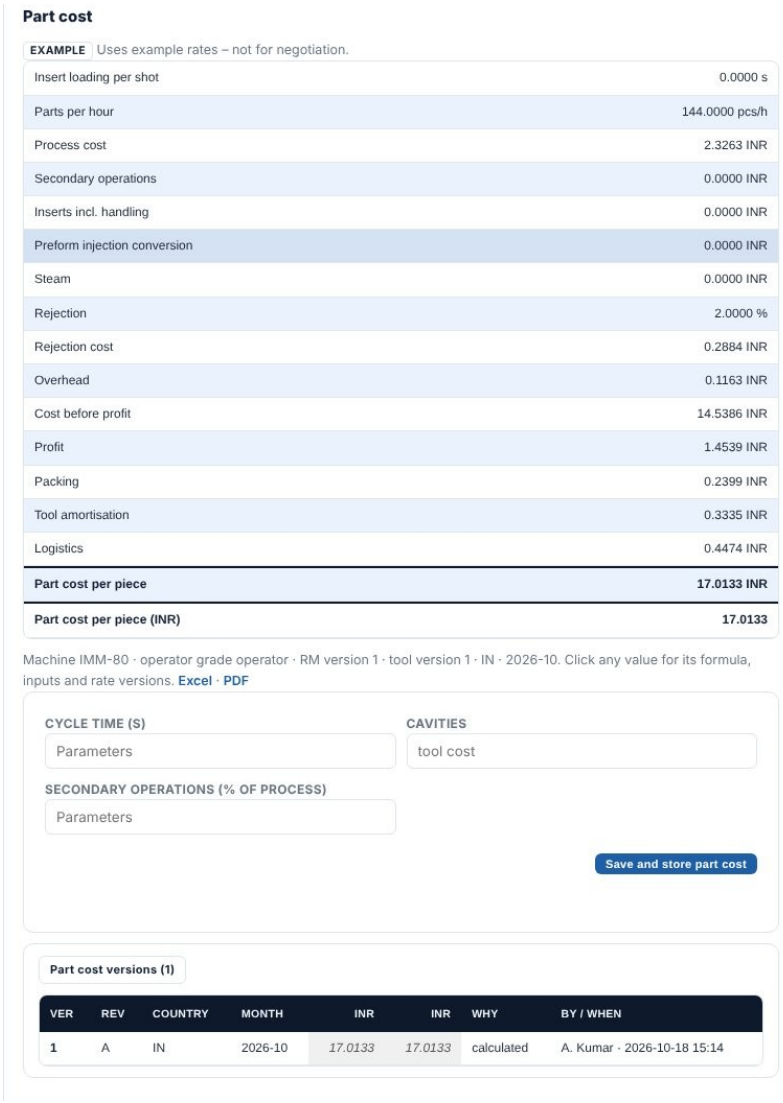
Mould / die concept

- A first concept: cavity layout, parting line, slides, ejection and gate.
- A starting point to discuss with the toolmaker.



Simulation check

- Fast check of draft angle, wall thickness, sink risk and cycle time.
- Corrections are listed with their area, so you know what to change in the design.



Part cost

EXAMPLE Uses example rates – not for negotiation.

Insert loading per shot	0.0000 s
Parts per hour	144.0000 pcs/h
Process cost	2.3263 INR
Secondary operations	0.0000 INR
Inserts incl. handling	0.0000 INR
Preform injection conversion	0.0000 INR
Steam	0.0000 INR
Rejection	2.0000 %
Rejection cost	0.2884 INR
Overhead	0.1163 INR
Cost before profit	14.5386 INR
Profit	1.4539 INR
Packing	0.2399 INR
Tool amortisation	0.3335 INR
Logistics	0.4474 INR
Part cost per piece	**17.0133 INR**
Part cost per piece (INR)	**17.0133**

Machine IMM-80 · operator grade operator · RM version 1 · tool version 1 · IN · 2026-10. Click any value for its formula, inputs and rate versions. Excel · PDF

CYCLE TIME (S)

Parameters

CAVITIES

tool cost

SECONDARY OPERATIONS (% OF PROCESS)

Parameters

Save and store part cost

Part cost versions (1)

VER	REV	COUNTRY	MONTH	INR	INR	WHY	BY / WHEN
1	A	IN	2026-10	*17.0133*	*17.0133*	calculated	A. Kumar · 2026-10-18 15:14

Part cost

- Raw material + machine hour + labour + overheads + tool amortisation.
- Saved as a numbered cost version. Compare versions at any time.



COUNTRY	EX-WORKS (LOCAL)	USD	EX-WORKS INR	FREIGHT+INS.	BCD+SWS	IGST	IGST CRE
IN	***16.57 INR***	***0.20***	***16.57***	***0.00***	***0.00 domestic: no import duty***	***0.00***	***0.***
CN	*1.51 CNY*	*0.21*	*17.52*	*0.76*	*2.01 standard rate*	*3.65*	*3.*
KR	*406.46 KRW*	*0.30*	*24.79*	*1.14*	*0.00 FTA preferential rate*	*4.67*	*4.*
JP	*45.39 JPY*	*0.30*	*25.42*	*1.25*	*0.00 FTA preferential rate*	*4.80*	*4.*
TW	*8.19 TWD*	*0.25*	*21.28*	*0.95*	*2.45 standard rate*	*4.44*	*4.*
TH	*7.53 THB*	*0.21*	*17.69*	*0.88*	*1.02 FTA preferential rate*	*3.53*	*3.*
VN	*5,069.01 VND*	*0.20*	*16.73*	*0.87*	*0.97 FTA preferential rate*	*3.34*	*3.*
DE	*0.37 EUR*	*0.40*	*33.79*	*1.86*	*3.92 standard rate*	*7.12*	*7.*
US	*0.37 USD*	*0.37*	*31.12*	*2.40*	*3.69 standard rate*	*6.70*	*6.*

Part and tool made in each country with that country's approved rates; the landed cost adds freight, insurance, duty, IGST (credit shown apart), clearance and inland transport in India. Cheapest tool landed: IN. **Excel** · **PDF**

Country comparison

- Same part, produced in different countries, using each country's approved rates.
- Landed cost can include freight and duty.



Revisions

COMPARE	REV	FILE	WHAT CHANGED	SOURCE	VOLUME (CM³)	BY / WHEN
☐	A (current)	plastic_cover.step		plastic_cover.step / plastic_cover	53.772	A. Kumar · 2026-10-18 15:14

Upload a new revision

STEP FILE

Choose File | No file chosen

REV NO. (BLANK = NEXT)

WHAT CHANGED

Upload revision

Where used

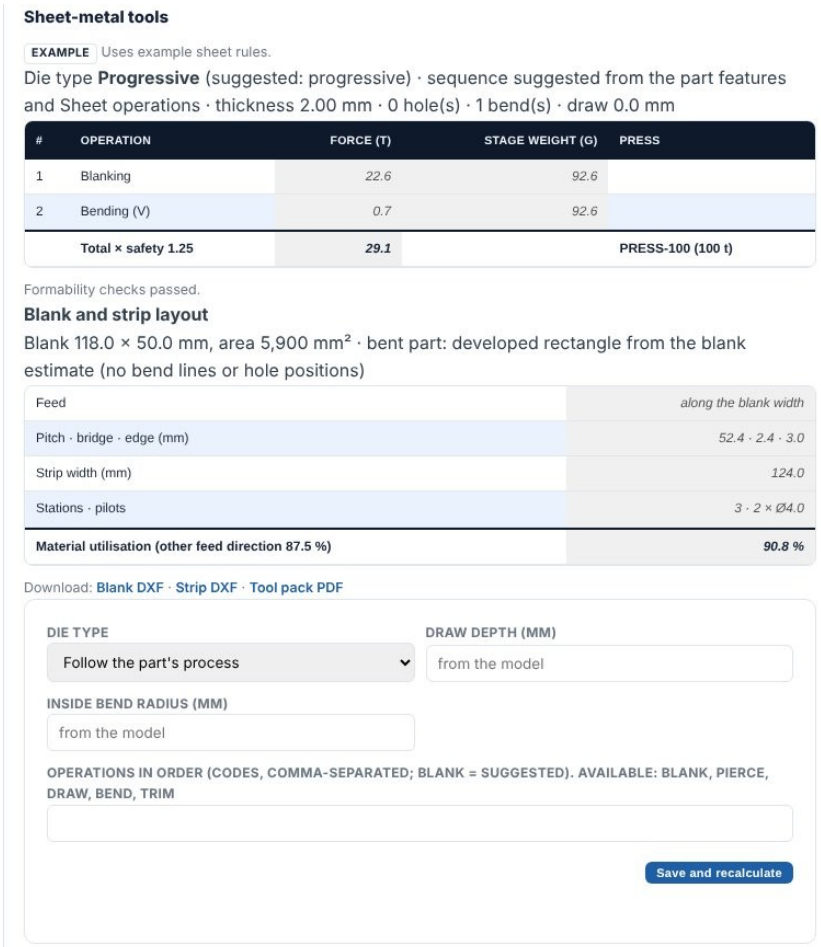
VARIANT	PARENT	QTY
STD	top level	1.00

History

WHEN	WHO	WHAT	DETAILS
2026-10-18 15:14	A. Kumar	status changed	old: concept_tool new: simulation comment:
2026-10-18 15:14	A. Kumar	updated	status: concept_tool → **simulation**
2026-10-18 15:14	A. Kumar	status changed	old: uploaded new: concept_tool comment:
2026-10-18 15:14	A. Kumar	updated	status: uploaded → **concept_tool**
2026-10-18 15:14	A. Kumar	piece cost_calculated	version: 1 part cost (INR): 17.0133 machine: IMM-80 country: IN costing month: 2026-10 reason: calculated rate versions: #9, #10, #11, #12, #13, #14, #15, #16, #17, #18, #19, #20, #21, #45, #46, #635, #1626
2026-10-18 15:14	A. Kumar	tool cost_calculated	version: 1 tool cost per tool (INR): 166766.31 tool CAPEX (INR): 166766.31 cavities: 1 tools: 1 tool country: IN costing month: 2026-10 reason: imported from a 3D file rate versions: #9, #10, #11, #12, #14, #15, #16, #17, #18, #19, #20, #22, #23, #24, #27, #28, #35, #36, #37, #38, #39, #40, #41, #42, #43, #44, #45, #46, #551, #594, #617, #635, #1341, #1342, #1343, #1344
2026-10-18 15:14	A. Kumar	rm cost_calculated	version: 1 net RM cost (INR): 11.8076 material: ABS country: IN costing month: 2026-10 reason: imported from a 3D file rate versions: #551, #552
2026-10-18 15:14	A. Kumar	revision added	rev: A file: plastic_cover.step change_note: source: plastic_cover.step / plastic_cover
2026-10-18 15:14	A. Kumar	created	part_code: SCT26-COVER sourcing: make status: uploaded name: SCT26-COVER material: ABS process: injection_moulding

Revisions and history

- Who changed what and when. Nothing is deleted.



Sheet-metal tools

- For brackets and stampings: progressive, compound or single-stage die layout and tool cost.

6.5 Rates: machine hour, library and approval

PRODUCING COUNTRY: India (INR) | COSTING MONTH (AS OF): October 2026 | Machine & labour rates | Material & process rates | Rate library | Rate review (1)

Uses example rates – not for negotiation
81 of the rate versions used for India are still EXAMPLE values. Replace them in the Rate library.

Machine hour rates (machine only, labour added per process)

MACHINE	PROCESS	DEPRECIATION (INR/H)	INTEREST (INR/H)	POWER (INR/H)	MAINTENANCE (INR/H)	FLOOR SPACE (INR/H)	CONSUMABLES (INR/H)	OVERHEAD (INR/H)	MHR (INR/H)	MHR (INR/H)
Band saw • BANDSAW	Raw material cutting	14.84	8.20	15.30	4.69	6.56	40.00	13.44	103.04	103.04
CNC vertical machining centre 850 • VMC-850	CNC machining	105.47	64.45	76.50	46.88	13.13	150.00	68.46	524.89	524.89
5-axis high-speed machining centre • HSM-5AX	CNC machining	488.28	358.07	140.25	260.42	21.88	400.00	250.33	1,919.23	1,919.23
CNC lathe • CNC-LATHE	CNC turning	70.31	42.97	51.43	31.25	8.75	100.00	45.71	350.41	350.41
Wire EDM • WEDM	Wire cut	140.63	85.94	59.50	78.13	8.75	250.00	93.44	716.38	716.38
Die-sinking EDM • EDM-SINK	EDM	93.75	57.29	61.20	52.08	8.75	180.00	67.96	521.04	521.04
Surface grinder • GRIND-SURF	Grinding	37.11	24.61	38.25	14.06	8.75	80.00	30.42	233.20	233.20
Deep-hole drilling machine • DRILL-DH	Deep-hole drilling	113.39	75.20	76.50	57.29	10.94	120.00	68.00	521.31	521.31
Polishing bench • POLISH	Polishing	6.51	2.60	6.38	1.04	4.38	60.00	12.14	93.04	93.04
Spotting / try-out press • SPOTTING	Assembly & spotting	54.69	50.13	19.13	18.23	16.41	30.00	28.29	216.86	216.86
Injection moulding machine 80 t • IMM-80	Injection moulding	39.06	35.81	76.50	26.04	10.94	50.00	35.75	274.10	274.10

Figure 10. Machine and labour rates (MHR) for the producing country and costing month.

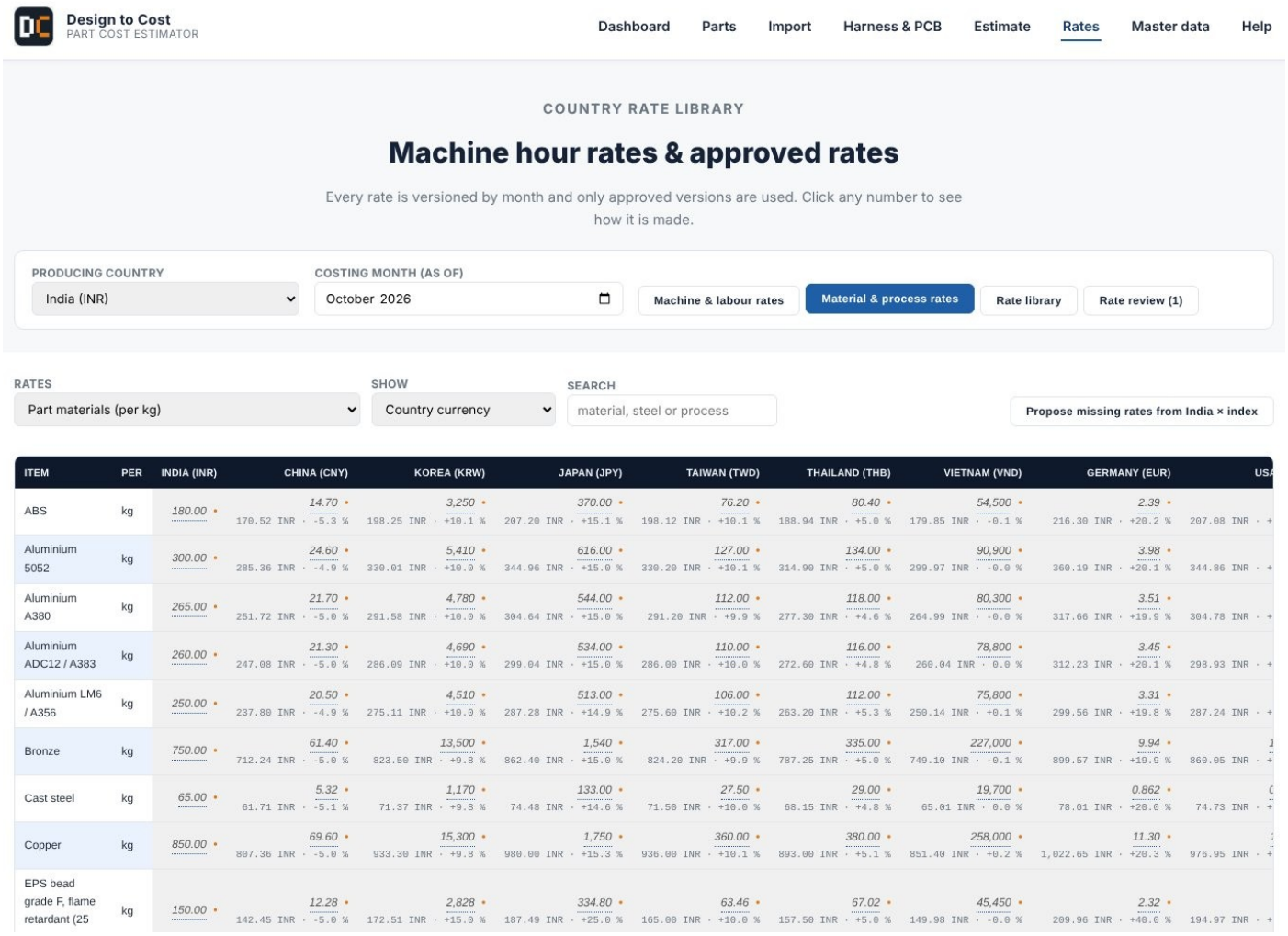


Figure 11. Material and process rates grid.

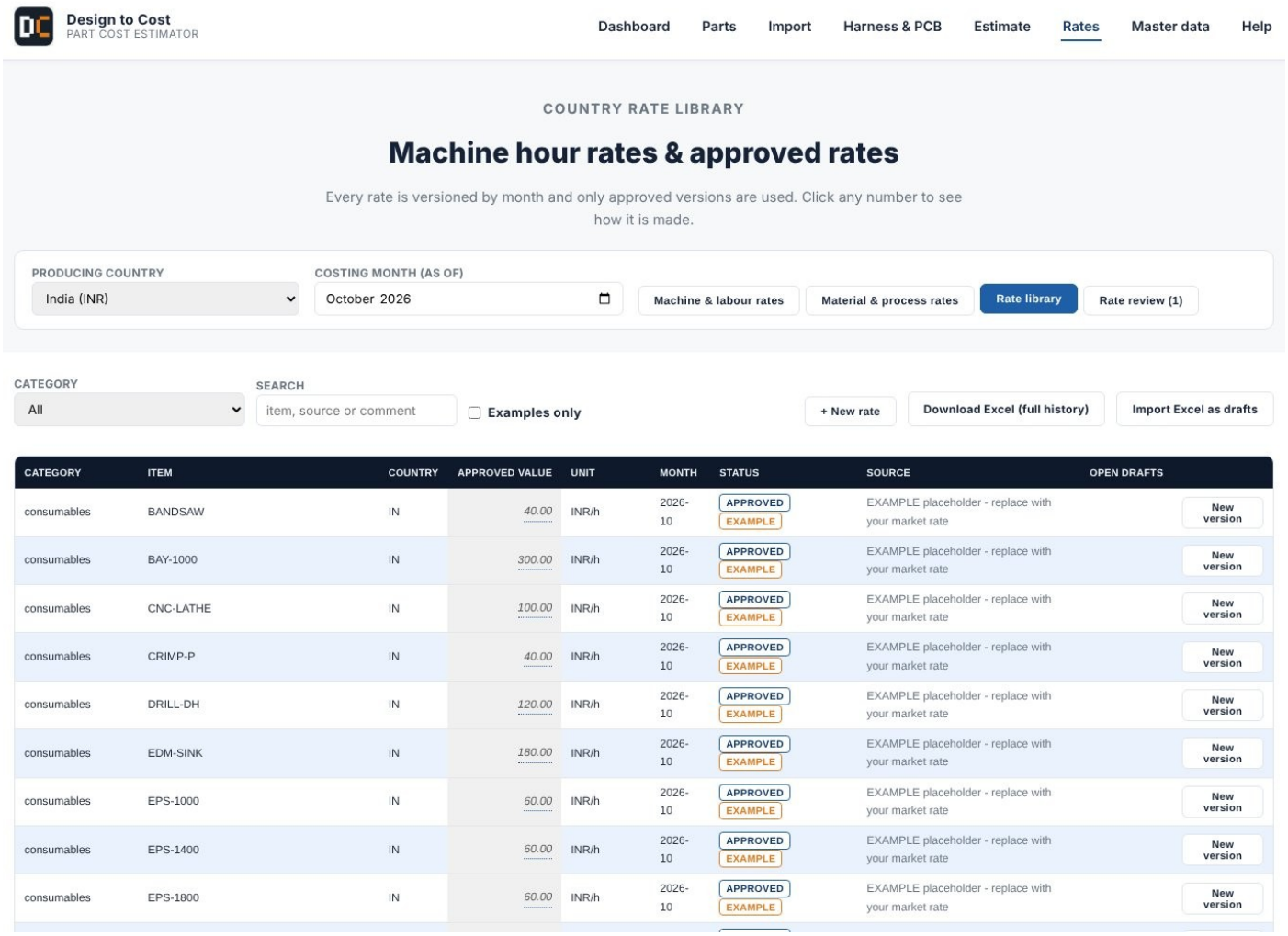


Figure 12. Rate library: every rate with status (Approved / Example), unit and month. Use New version to change a rate.

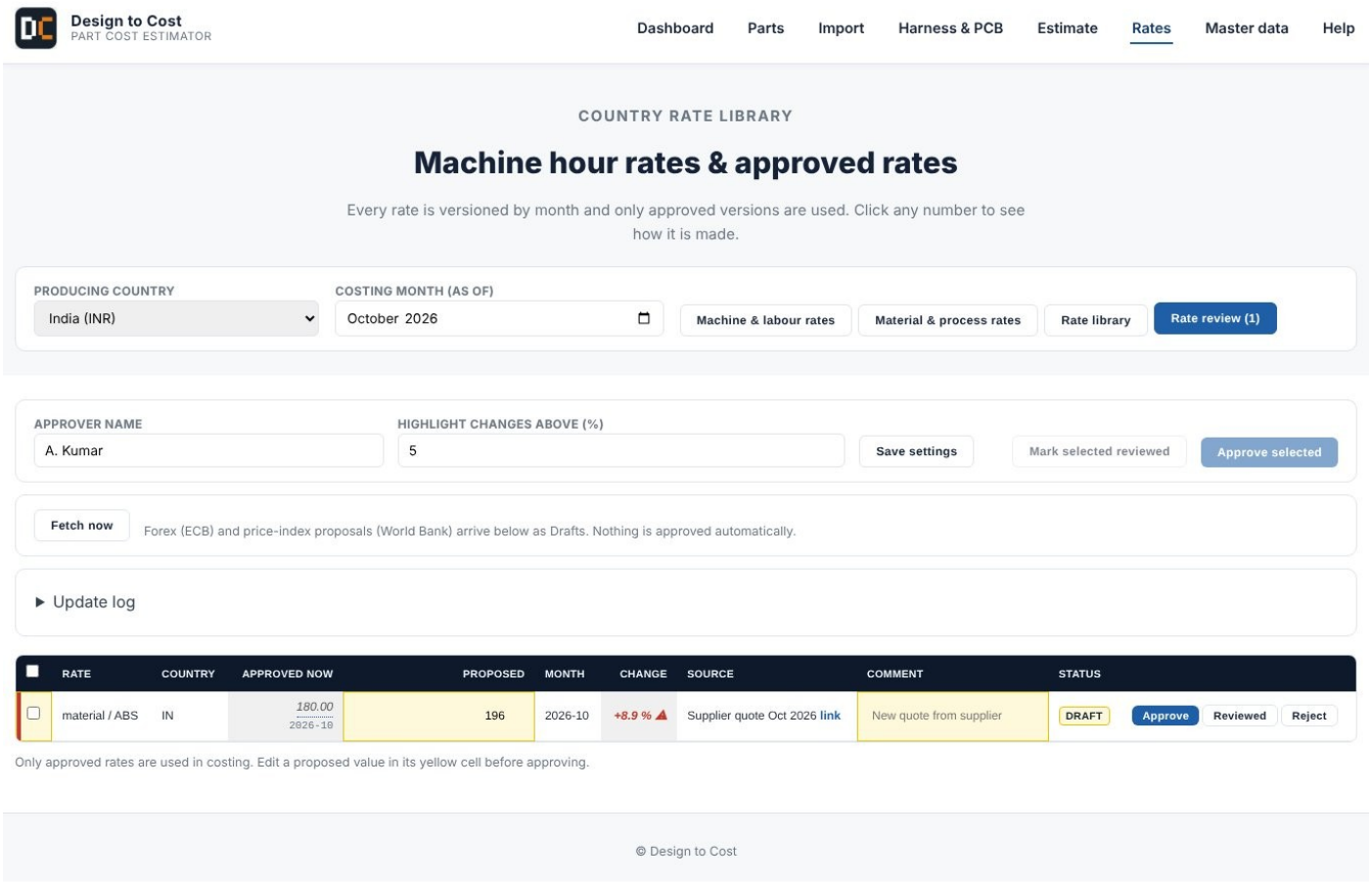


Figure 13. Rate review: new proposals arrive as Draft. A reviewer marks Reviewed, an approver clicks Approve. Only approved rates are used in costing.

6.6 Master data and replacing the example values

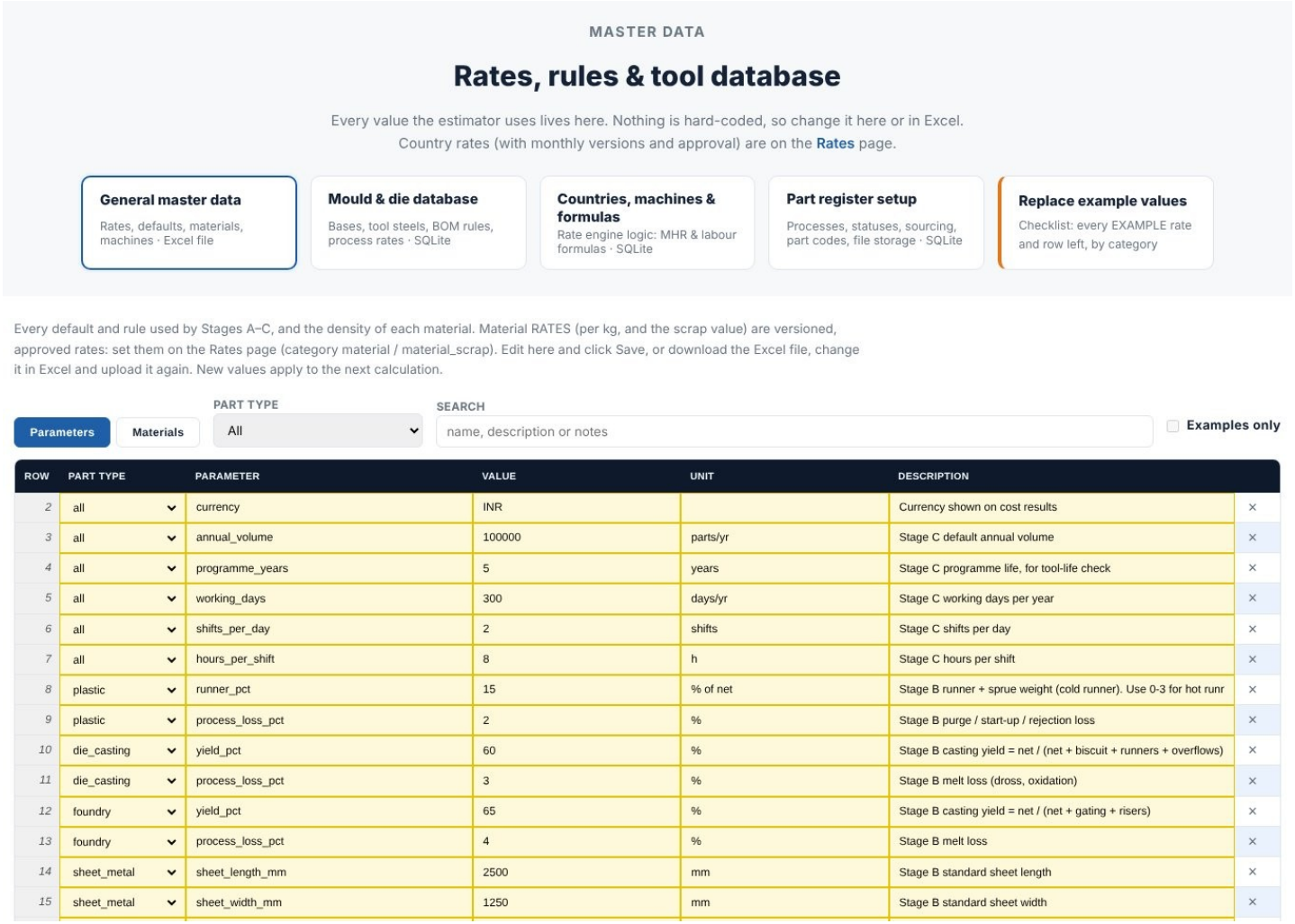


Figure 14. Master data: editable tables for rates, formulas and rules.

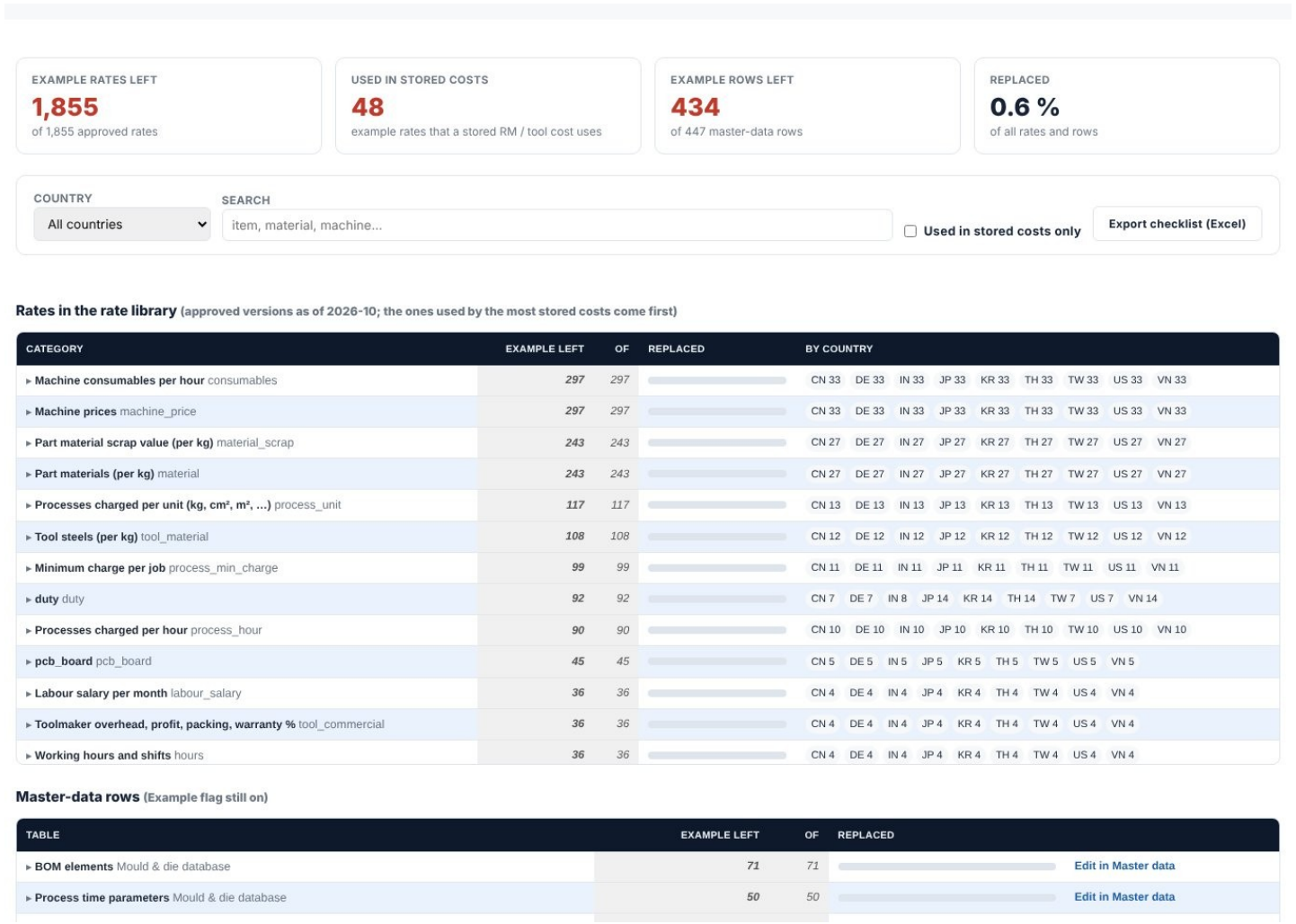


Figure 15. Replace example values: a checklist of every EXAMPLE value. Work down the list and enter your real rate; the red banner disappears when all are replaced.

6.7 Harness and PCB (electrical BOM)

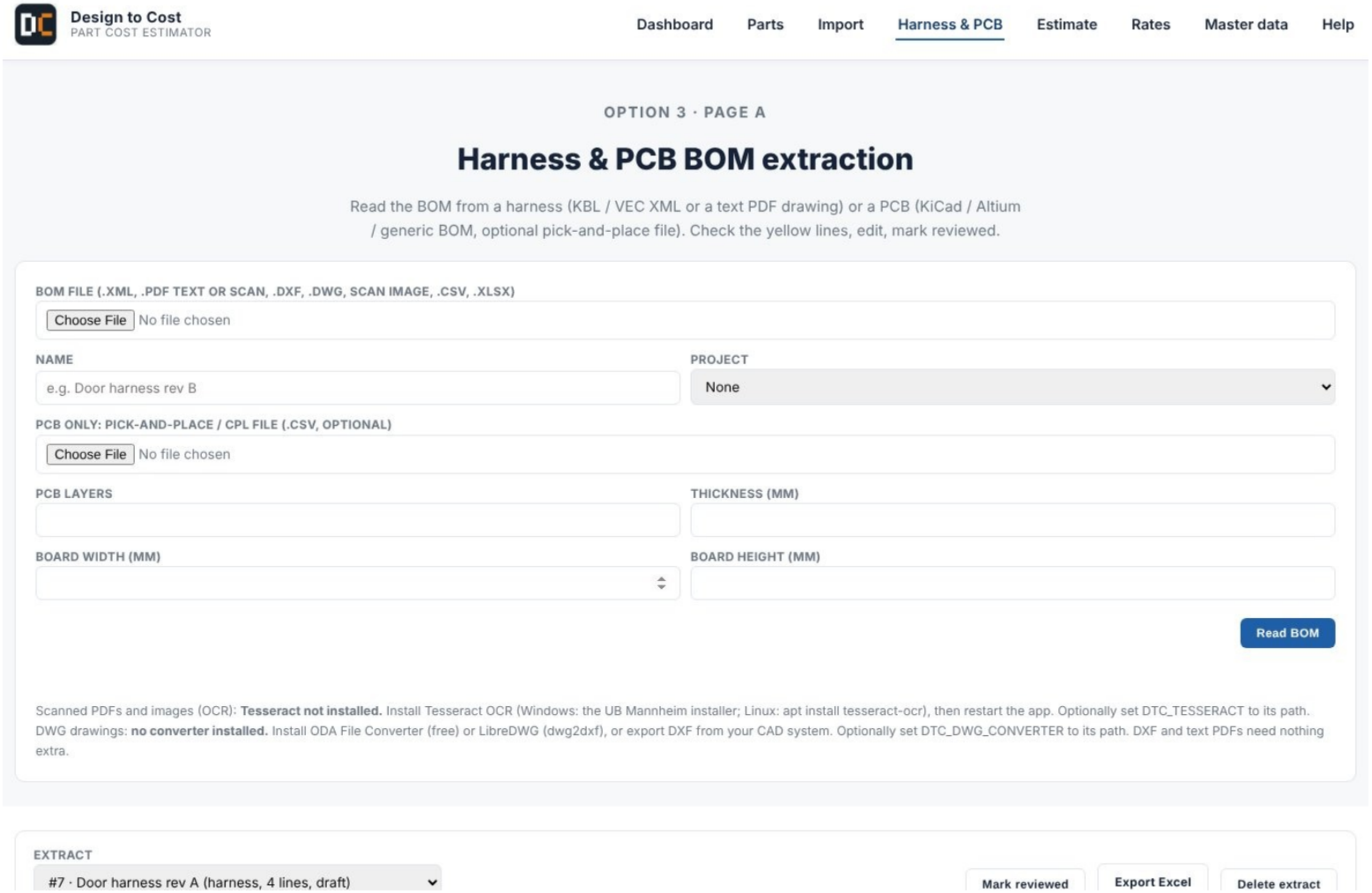


Figure 16. Harness & PCB page: upload a wiring harness (KBL/XML) or a BOM.

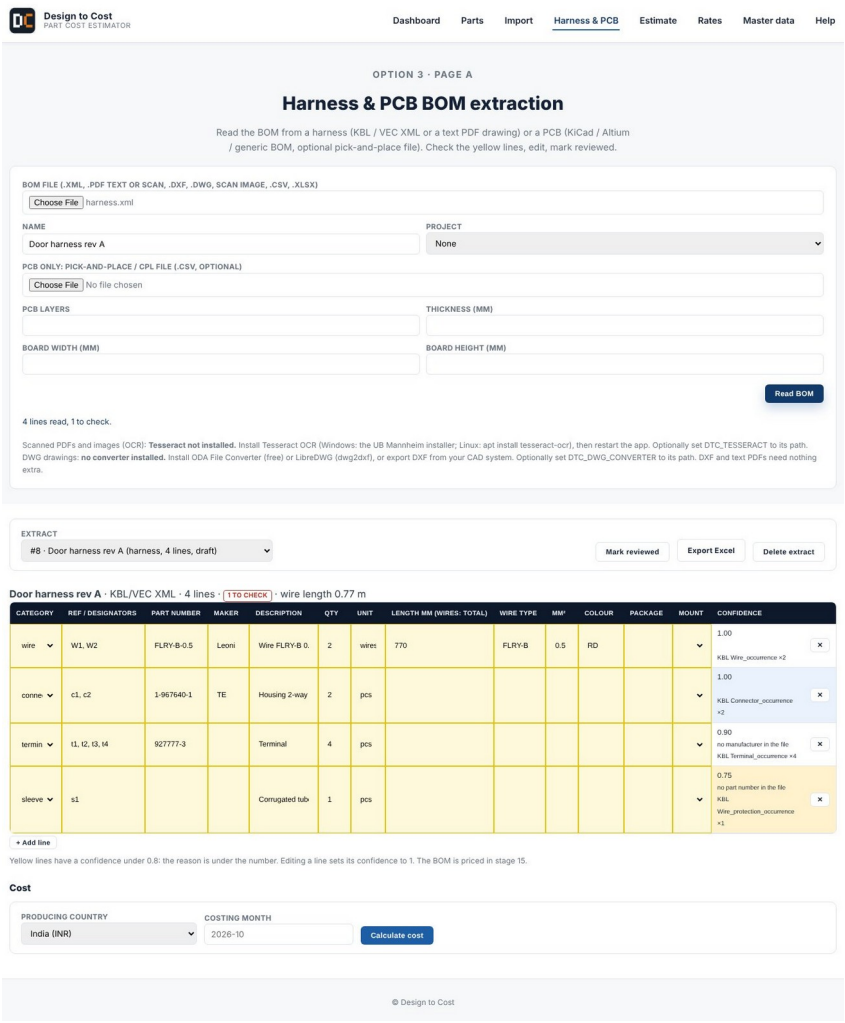


Figure 17. Review the lines read from the file: category, quantity, length and confidence.

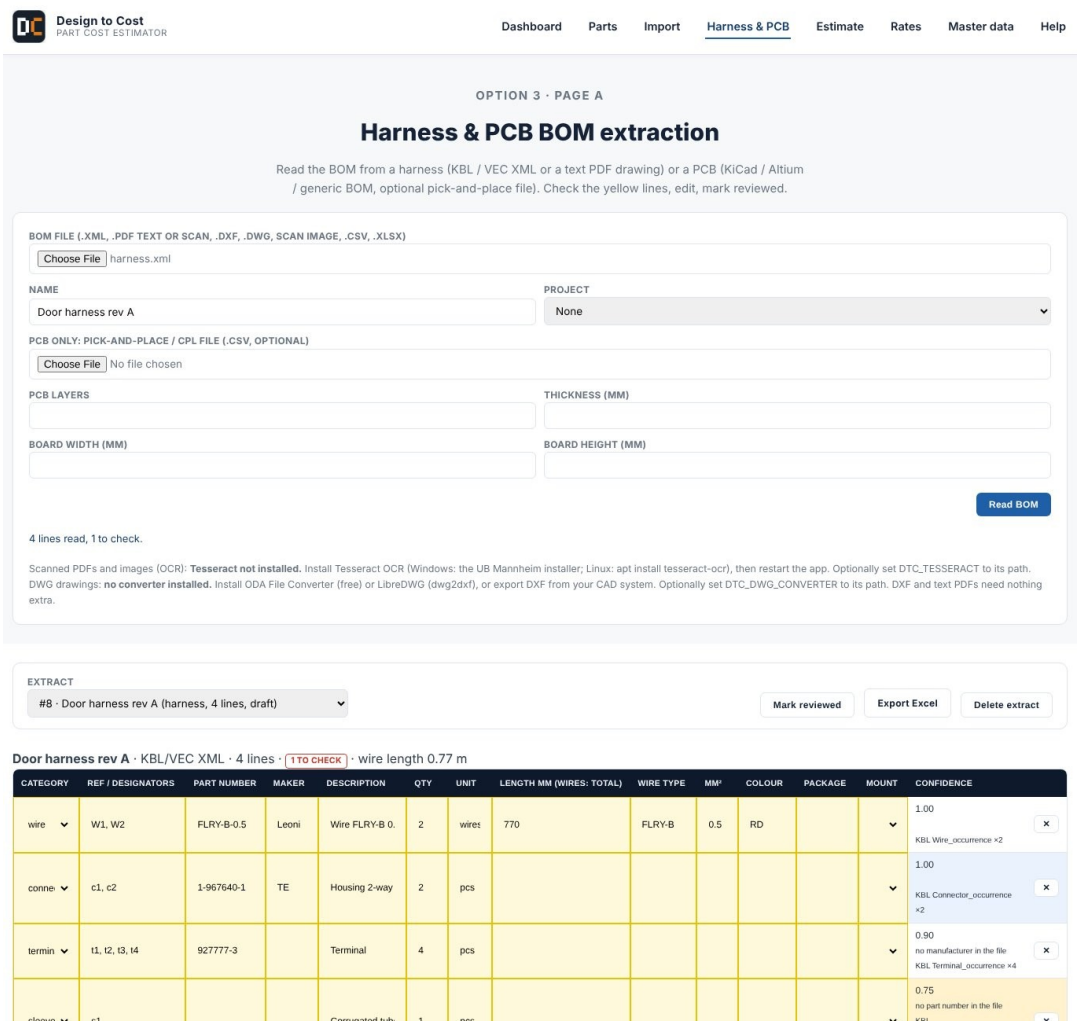


Figure 18. Cost of the harness or board, line by line.

6.8 Help

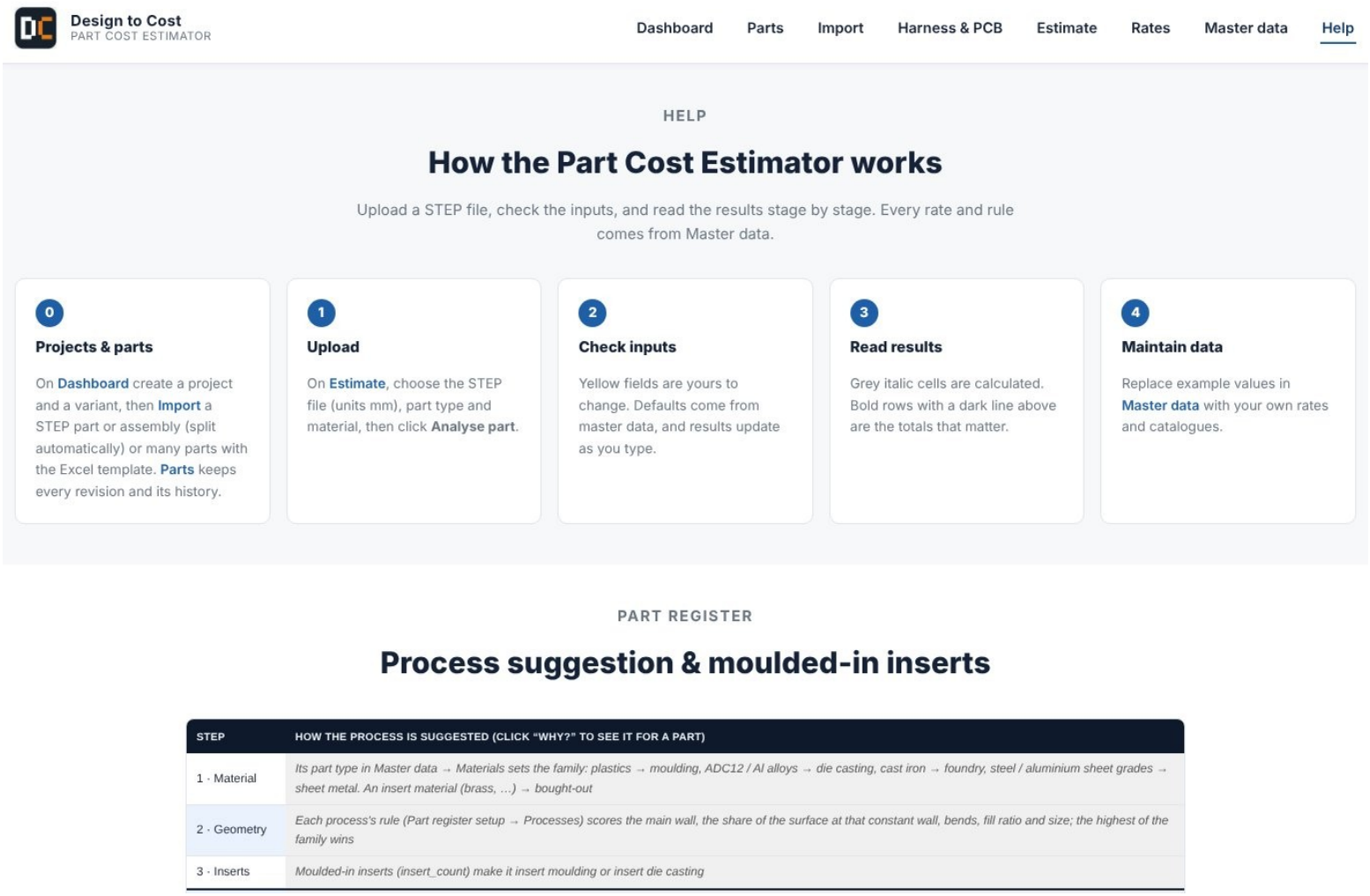


Figure 19. Built-in help page.

Other modules: EPS (expanded polystyrene) moulding with steam and thermocouple data, and PET stretch blow moulding (preform and bottle mould) are on the Estimate and part panels; select them as the process.

7. Replace the example values (do this first)

1. Open Master data, then Replace example values (or go to http://127.0.0.1:8000/master-data/examples).
2. Work through the list. For each item enter your real rate and source.
3. Open Rates, Rate review. Approve the new rates (name of approver is stored).
4. When the list is empty, the red 'example rates' banner is gone and costs can be shared.

8. Backup, update and corrections

Backup the whole working data (projects, rates, uploaded files) before any change:

```
python backup.py
python backup.py --restore latest
python backup.py --restore latest --only master_data
```

Update to the newest version of the software (your own rates and projects are kept):

```
python update.py
python update.py --keep-mine     (keep your edited files on a conflict)
python update.py --take-new      (take the new version on a conflict)
```

Corrections: re-download or update the code, restart the server. Cost versions already saved stay as they were; new calculations create a new version number.

9. Quick reference

I want to...	Go to
Get a quick part cost	Estimate
Create a project and parts	Import, then Parts
See cost vs target and CAPEX	Dashboard
Change a machine or material rate	Rates, Rate library, New version, then approve
See how a number is made	Click the number (traceability panel)
Cost a wiring harness or PCB	Harness & PCB
Save my work	python backup.py

Term	Meaning
MHR	Machine hour rate, cost of running a machine for one hour
BOM	Bill of materials
CAPEX	One-time investment, mainly tools and moulds
STEP	Neutral 3D CAD file format (.step / .stp)
EXAMPLE	Placeholder value that must be replaced with your own

10. Working in a team (server mode)

For more than one user, run the software on one PC or server in server mode. Users then log in with a name and password and have a role:

Role	May do
viewer	Look only
engineer	Create projects, parts, cost; propose rates as Draft

approver	Also approve rates
admin	Also manage users

```
set DTC_MODE=server
set DTC_ADMIN_PASSWORD=choose-a-long-password
uvicorn backend.app.main:app --host 0.0.0.0

(Mac/Linux: use export instead of set)
python -m backend.app.server_tools create-user alice --role engineer
```

Colleagues open http://<server-IP>:8000 and log in. For large teams use PostgreSQL by setting DATABASE_URL; see README.md.

11. Known limits

- Rates shipped with the tool are examples until you replace them.
- Sheet-metal parts get a tool cost but no full part cost yet (plastic and die-cast parts do).
- Scanned drawings need Tesseract installed; DWG files need a converter. DXF and Excel work directly.
- Simulation is a fast screening estimate, not a replacement for mould-flow software.
- Costs are estimates for planning and negotiation preparation. Confirm with supplier quotes.